

Car Price Analytics

Visual Dashboard & Insights Report

Dataset: 205 vehicles | 26 attributes | 22 brands

Analytics Dashboard

The dashboard below presents six visualisations derived from the 205-vehicle dataset, covering body style, dimensions, drive wheel type, horsepower, engine cylinders, and brand comparison.

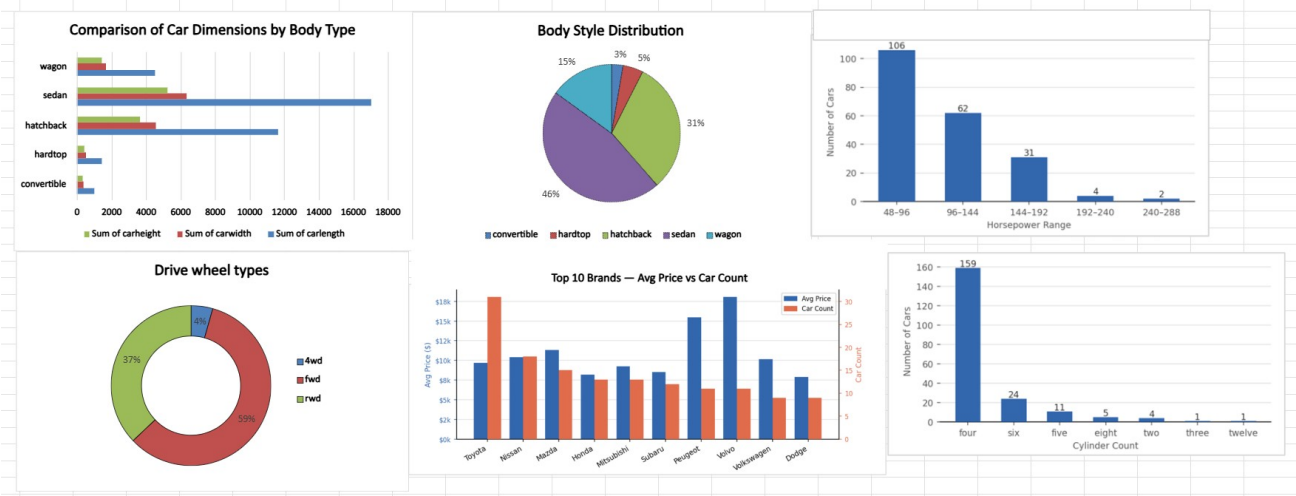


Figure 1 — Car Price Analytics Dashboard (Excel)

Chart-by-Chart Insights

1. Body Style Distribution (Pie Chart)

The pie chart reveals a market dominated by practical body styles:

- Sedans lead with 46% of the fleet — the go-to choice for everyday commuters and families.
- Hatchbacks hold 31%, forming a strong economy segment alongside sedans.
- Wagons account for 15%, serving utility-oriented buyers.
- Hardtops (5%) and convertibles (3%) are premium niche segments with low volume but high price points.

Key Takeaway

Sedans + hatchbacks = 77% of the market. Any pricing or inventory strategy must centre these two body types.

2. Comparison of Car Dimensions by Body Type (Bar Chart)

The horizontal bar chart compares car height, width, and length across body styles:

- Sedans are the longest vehicles overall — reflective of their spacious cabin design.
- Hatchbacks are shorter in length but comparable in width, making them ideal for urban driving.

- Wagons are notably taller, consistent with their cargo-priority design.
- Convertibles and hardtops are the most compact, prioritizing sporty aesthetics over space.

Key Takeaway

Sedan length (~187 in avg) gives buyers the most interior space; hatchbacks sacrifice length for maneuverability.

3. Drive Wheel Types (Donut Chart)

The donut chart shows the drivetrain split across the dataset:

- FWD (front-wheel drive) dominates at 59% — aligned with the dataset's economy-car orientation and lower manufacturing cost.
- RWD (rear-wheel drive) holds 37%, closely associated with performance and luxury segments (BMW, Jaguar, Porsche-class cars).
- 4WD is a small fraction at 4%, appearing mainly in Subaru and Audi models.

Key Takeaway

RWD vehicles skew significantly higher in price; FWD is the default for value-focused buyers. Drivetrain is a strong price predictor.

4. Horsepower Distribution (Bar Chart)

The histogram of horsepower shows a steep right-skewed distribution:

- 106 vehicles (52%) fall in the 48–96 HP range — firmly economy territory.
- 62 cars (30%) sit in the 96–144 HP range, covering mainstream family vehicles.
- Only 37 cars (18%) exceed 144 HP; truly high-performance vehicles (192+ HP) are rare at just 3%.

Key Takeaway

The dataset is overwhelmingly low-power. Performance cars are outliers — relevant to note for premium pricing models.

5. Engine Cylinder Count (Bar Chart)

The cylinder distribution mirrors the horsepower findings:

- 4-cylinder engines dominate at 159 vehicles (78%) — the standard for economy and mid-range cars worldwide.
- 6-cylinder engines appear in 24 cars (12%), bridging performance and comfort.
- 5-cylinder (11 vehicles), 8-cylinder (5), and exotic configurations (2, 3, 12 cylinders) are niche segments.

Key Takeaway

Engine cylinder count is a reliable proxy for price tier: 4-cyl = economy, 6-cyl = mid/performance, 8+ = premium/luxury.

6. Top 10 Brands — Average Price vs Car Count (Dual-Axis Bar Chart)

The dual-axis chart plots both average price (blue bars, left axis) and car count (orange bars, right axis):

- Toyota leads in volume (31 cars) but maintains a modest average price (~\$10K), reflecting its mass-market positioning.
- Volvo and Volkswagen are notable for lower volumes but higher average prices, indicating a premium-per-unit strategy.
- Nissan, Mazda, and Honda are high-volume, low-price brands — competitive economy players.
- Dodge shows a mid-range average price with moderate volume; Peugeot and Mitsubishi are similarly positioned.

Key Takeaway

Volume and price are inversely correlated for most brands. Toyota's scale does not command premium pricing; Volvo's scarcity does.

Summary of Findings

Across all six visualisations, five consistent themes emerge:

- Economy dominates: Most cars are small, 4-cylinder, FWD vehicles priced under \$15,000.
- Sedans are pivotal: Highest volume and broadest price range — the most strategically important segment.
- Premium is niche: Convertibles, hardtops, RWD, 6+ cylinders, and high-HP cars are rare but high-value.
- Japanese brands lead by volume: Toyota alone represents 15% of the market; the top 6 brands are all Japanese.
- Price predictors: Body style, drivetrain, and cylinder count are the three strongest indicators of vehicle price.

End of Report — Based on CarPrice_Assignment.xlsx | 205 Records